Lecture 10

# Measurements and Electronic Instrumentation

**Slide Courtesy:**

1. Principles and Applications of Electrical Engineering, Giorgio Rizzoni,James Kearns

# Outline

- Understand the principles of measurement systems.
- Components of measurement systems.
- Wheatstone's Bridge

# Measurements and Electronic Instrumentation

**Measurement:** The process of quantifying physical quantities (e.g.,voltage, temperature, pressure) using standardized units.

**Electronic Instrumentation:** Devices and systems that measure, process, and display these quantities electronically.

# Types of Measurements

- **Electrical Measurements:**
  - Voltage, Current, Resistance, Capacitance, Inductance.
- **Physical Measurements:**
  - Temperature, Pressure, Displacement, Speed.
- **Chemical Measurements:**
  - pH, Concentration of gases.

# Measurement Instruments

**Name a few**

- Multimeter
- Oscilloscope
- Thermometer

# Key characteristics of measurement system

- **Accuracy:** How close a measured value is to the true value.
- **Precision:** Consistency of repeated measurements.
- **Resolution:** Smallest detectable change in the measured value

- **Error:** Difference between measurement and true value.
  - **Systematic**: Consistent, repeatable errors (e.g., calibration issues).
  - **Random**: Unpredictable variations (e.g., noise in circuits).
- **Span:** Linear operating range
- **Range**: The range of measurable values



High Precision, Low Accuracy



Low Precision, Good Accuracy



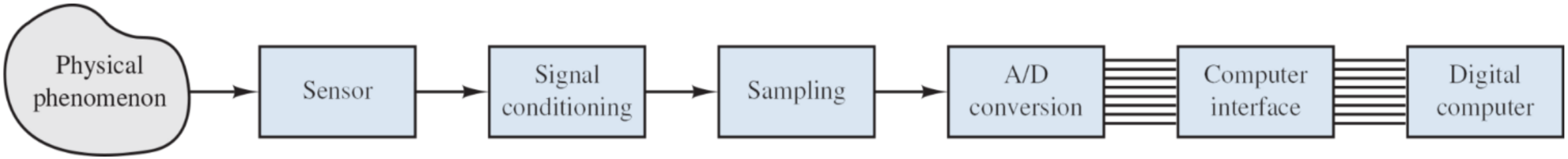
Low Precision, Low Accuracy



High Precision, High Accuracy

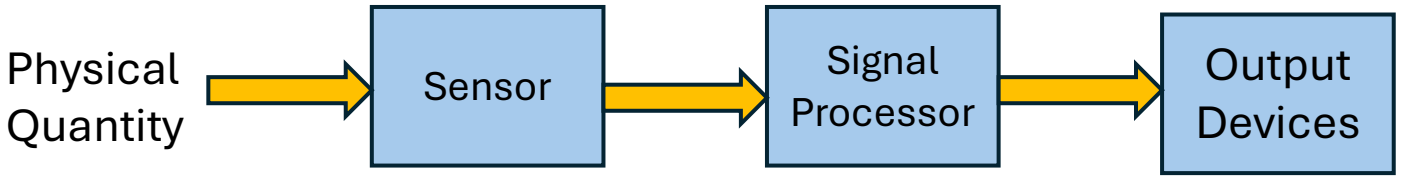
# Measurement Process

1. **Sensing**: Detecting the physical quantity (e.g., thermocouple for temperature).
2. **Signal Conditioning**: Amplifying, filtering, or converting the signal (e.g., analog-to-digital conversion).
3. **Data Processing**: Analyzing and interpreting the signal.
4. **Output**: Displaying or recording the result (e.g., LCD, computer).




Measurement Process

# Components of Electronic Instrumentation



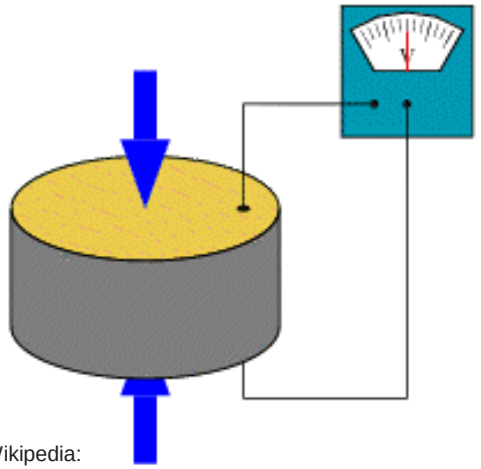

**1.Sensors and Transducer**: Converts physical parameters to electrical signals.

**Examples**:

- Thermocouple (temperature → voltage).
- Strain gauge (mechanical strain → resistance change).
- Photodiode (light intensity → current).

•**Characteristics**:

- Sensitivity: Output change per unit input.
- Linearity: Proportionality between input and output.
- Range: Minimum and maximum measurable values.



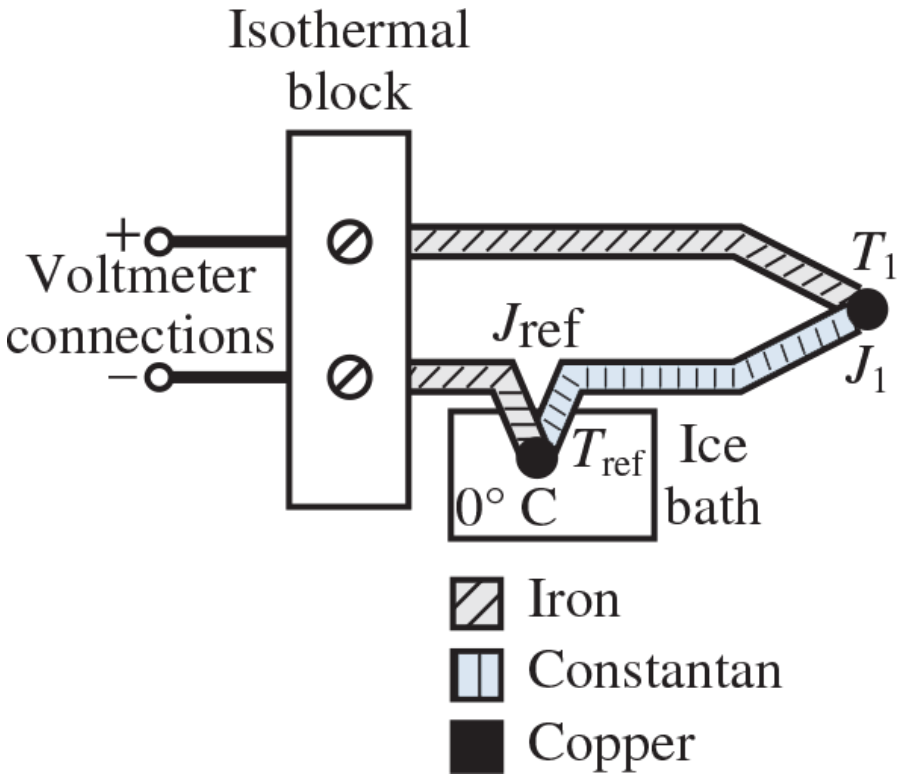
PC: Wikipedia:
A **piezoelectric disk** generates a voltage when deformed (change in shape is greatly exaggerated)

# ❑ Sensor example: Temperature Measurements

Two common temperature sensor types:

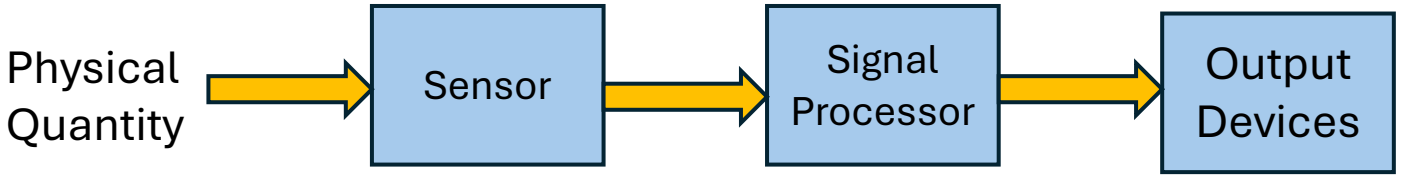
**Thermocouple:** Used to measure temperature by converting thermal energy into electrical energy.

- It operates based on the **Seebeck Effect**, when two different metals are joined together, a voltage (thermoelectric EMF) is generated at the junction due to the temperature difference between the two ends.
- **Hot Junction (J1)**:The point where the two metals are joined together and placed at the temperature to be measured.
- **Cold Junction (Reference Junction)**: maintained at a known reference temperature
- The voltage generated is proportional to the temperature difference between the hot and cold junctions.




- **Seebeck Effect:** Different metals have distinct electron densities and thermal conductivities, resulting in different rates at which electrons diffuse from the hot junction to the cold junction.
- This difference in electron diffusion creates a measurable voltage.

# Components of Electronic Instrumentation ..



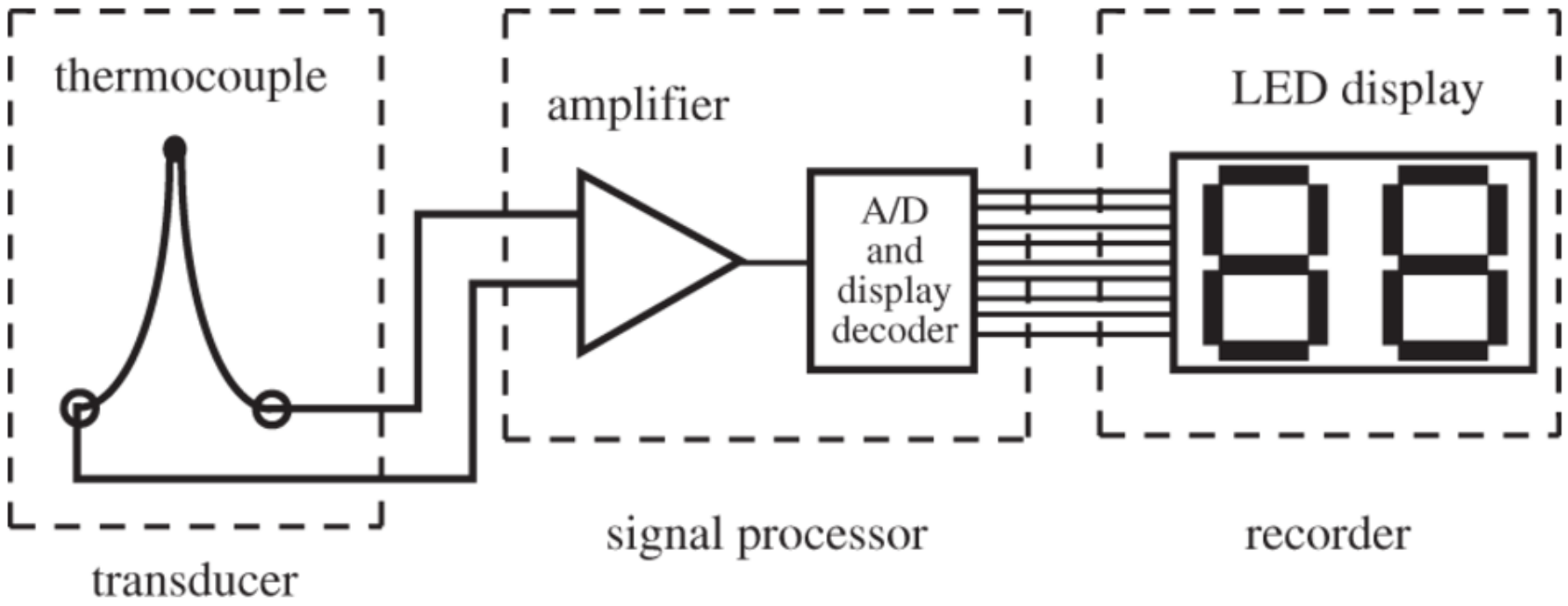

**2. Signal Conditioning Circuits**

- **Purpose**: Prepare sensor signals for processing.
- **Common Operations**:
  - **Amplification**: Boost weak signals (e.g., operational amplifiers).
  - **Filtering**: Remove noise (e.g., low-pass filters).
  - **Linearization**: Correct non-linear sensor outputs.
  - **Analog-to-Digital Conversion (ADC)**: Convert analog signals to digital for processing.

**3. Output Devices**

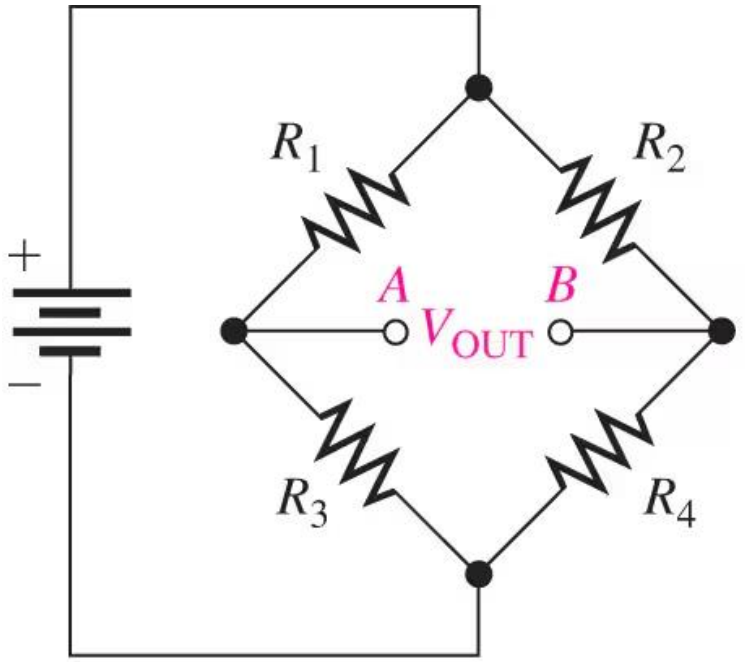
- **Displays**: LCDs, LEDs, or computer screens.
- **Recorders**: Data loggers or storage devices.
- **Actuators**: Devices that act on the measured data (e.g., relays, motors).

# Measurement System – Digital Thermometer



# Wheatstone's Bridge: A Fundamental Tool in Electrical Measurements

- Circuit used to measure unknown resistance precisely
- It is a fundamental tool in electrical and electronic measurement systems due to its high accuracy and precision.
- It is commonly used in applications like strain gauges, load cells, pressure sensors, temperature sensors (thermistors and RTDs), and other sensors that rely on **changes in resistance** to measure physical quantities like **force, pressure, or temperature**.

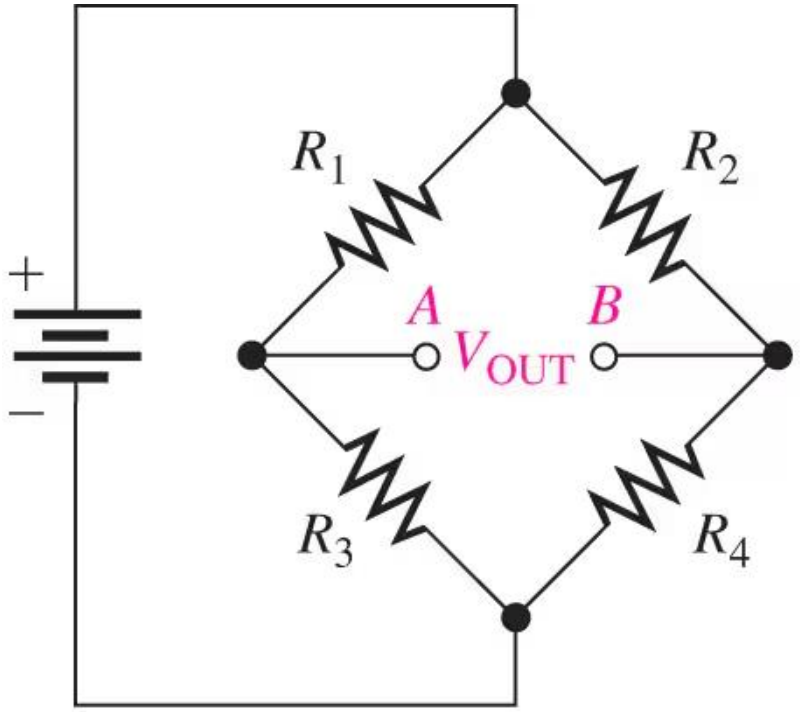


# Wheatstone's Bridge: A Fundamental Tool in Electrical Measurements

**Balanced bridge condition:**

- when the output voltage between terminals A and B is **equal to zero** ($V_{out} = 0V$)
- Also, the products of resistances in the opposite diagonal arms are equal. ($R_1/R_3 = R_2/R_4$)

**An unknown resistance in one position can be accurately measured using the balanced state of the bridge.**

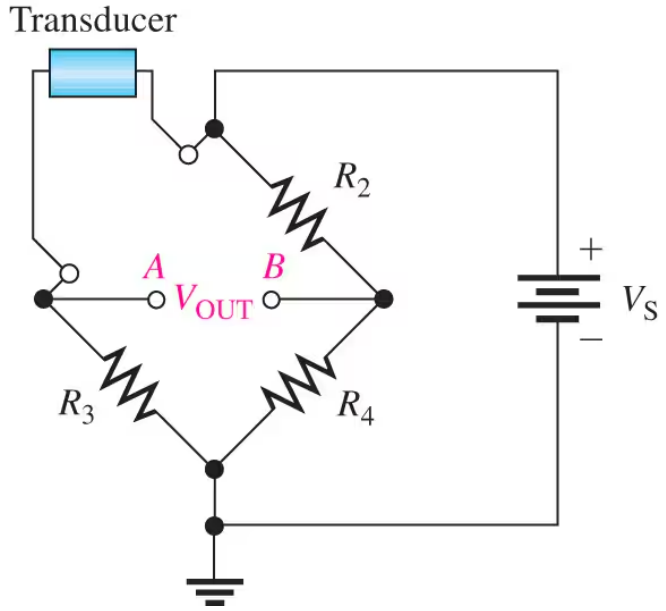


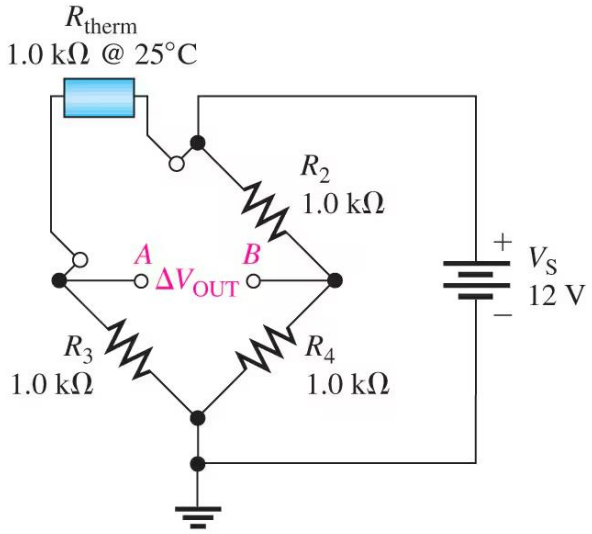

**Unbalanced State:**

- A bridge circuit that is in the **unbalanced state** as indicated by a voltage across the bridge that is proportional to the amount of deviation from the balanced state.

# A Bridge Circuit for Measuring physical parameter

- The resistance of the transducer changes proportionally to the changes in the parameter that it is measuring.
- If the bridge is balanced at a known point, then the amount of **deviation from the balanced condition**, as indicated by **the output voltage**, indicates the amount of change in the parameter being measured. Therefore, the **amount of imbalance** can determine the **value of the parameter being** measured.



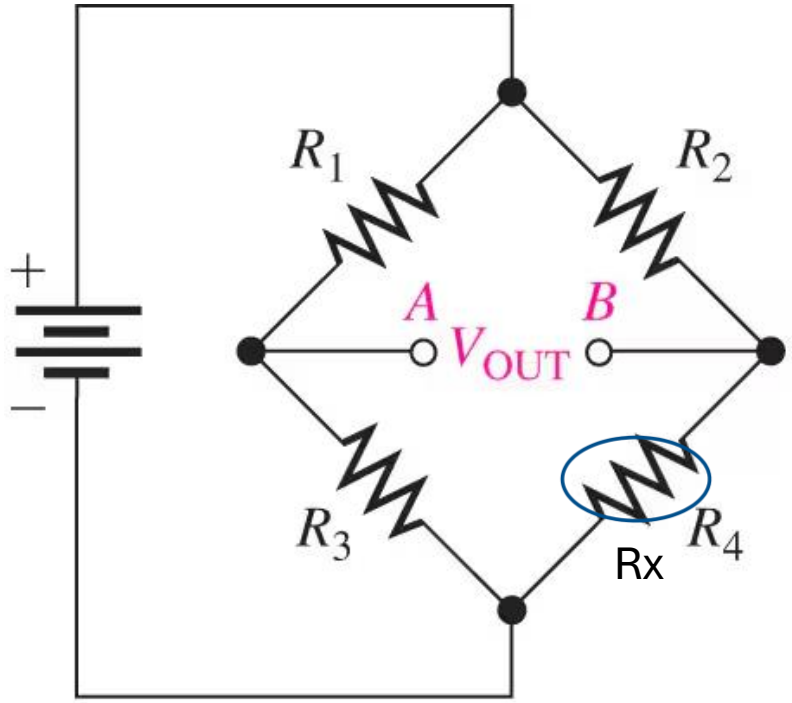

**A Bridge Circuit for Measuring Temperature:**
It is balanced at a reference temperature and becomes unbalanced at a measured temperature. For example, let's say the bridge is to be balanced at 25° C . A thermistor will have a known value of resistance at 25° C .

# Example : Balanced Bridge

Given R1 = 100 Ω, R2 = 150 Ω, R3 = 200 Ω, find Rx at **balance.**

**Balanced bridge condition:**

$R_1/R_3 = R_2/R_4$



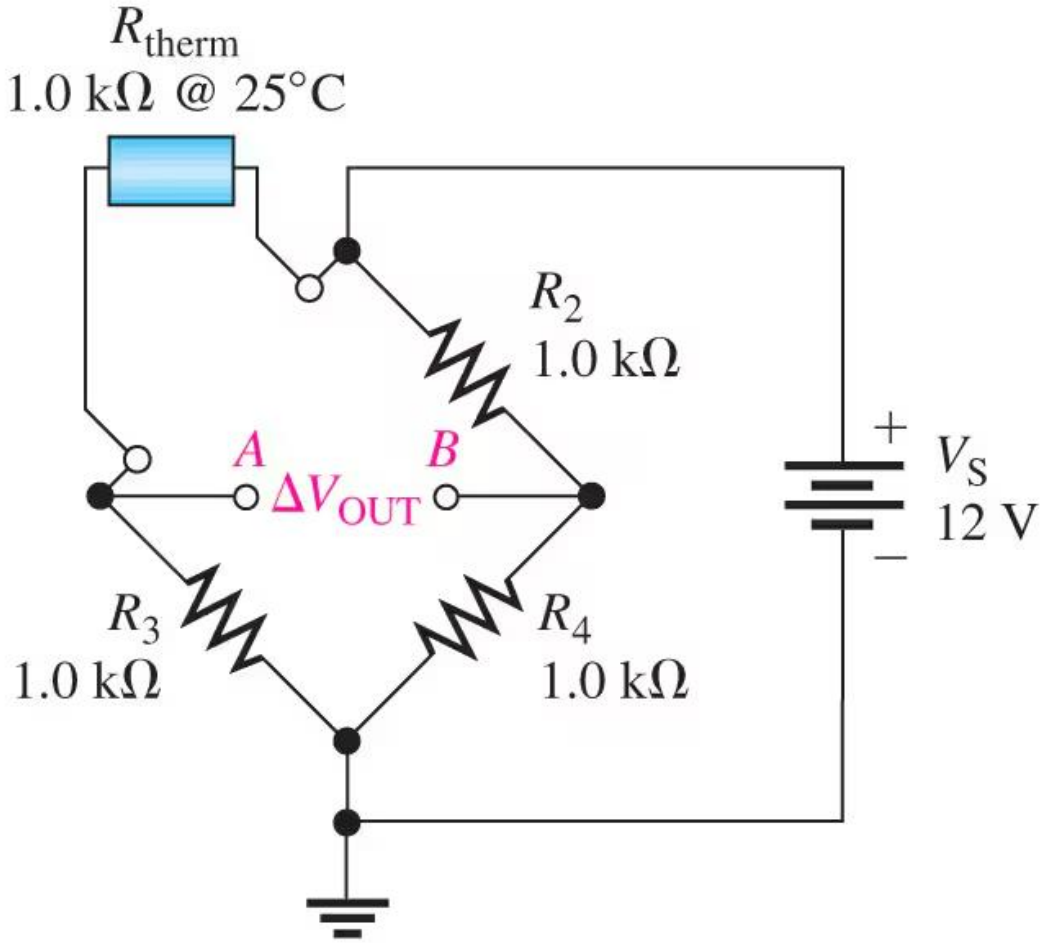

**Ans:**

100/200 = 0.5 and 150/300 = 0.5

Rx = $R_4$ = 300 Ω

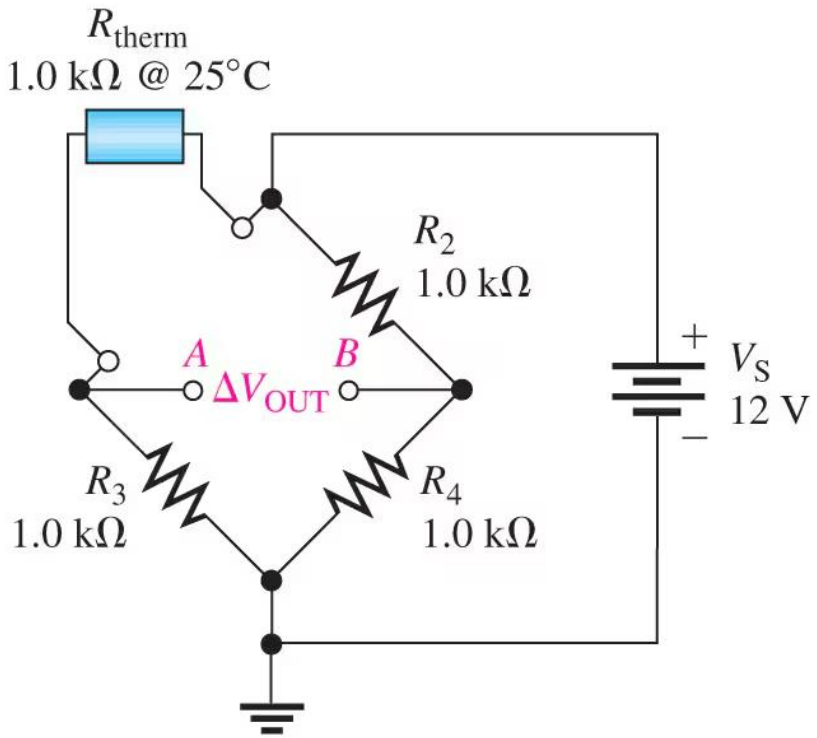
# Example: Unbalanced Bridge

Q1. Determine the output voltage of the temperature-measuring bridge circuit if the thermistor is exposed to a temperature of 50° C and its resistance at 25° C is 1k Ω . Assume the resistance of the thermistor **decrease**s to **900 Ω at 50° C** .




At 25° C is 1k Ω

**At 900 Ω at 50° C**




Apply the voltage-divider formula to the left side of the bridge at 50°C.

$$V_A = \left(\frac{R_3}{R_3 + R_{\text{therm}}}\right) V_S = \left(\frac{1.0\text{k}\Omega}{1.0\text{k}\Omega + 900\Omega}\right) 12\text{V} = 6.32\text{V}$$

Apply the voltage-divider formula to the right side of the bridge.

$$V_B = \left(\frac{R_2}{R_2 + R_4}\right) V_S = \left(\frac{1.0\text{k}\Omega}{2.0\text{k}\Omega}\right) 12\text{V} = 6.00\text{V}$$

The output voltage at 50°C is the difference between $V_A$ and $V_B$.

$$V_{\text{OUT}} = V_A - V_B = 6.32\text{V} - 6.00\text{V} = \mathbf{0.32V}$$

Node *A* is positive with respect to node *B*.